

Shaheer Khalid

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EDUCATION

Forman Christian College University

Lahore, Pakistan

BSc (Hons) Computer Science

Expected December 2027

Relevant Courses: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Software Engineering, Discrete Mathematics

EXPERIENCE

Forman Christian College University

Lahore, Pakistan

Teaching Assistant, Computer Science

Sep 2025 – Present

- Mentor students in Python, algorithms, and OOP; write and grade quizzes and lab exercises covering system design and database fundamentals.
- Conduct viva exams to test reasoning depth, not just answer recall.

FCSC – Forman Computer Science Club

Lahore, Pakistan

Director of Operations

Sep 2025 – Present

- Run club operations and coordinate workshops, hackathon teams, and technical speaker events for the FCCU CS community.

PROJECTS

UniPool – University Carpooling App

Software Engineering Course Project | FCCU

- Built a carpooling platform for day-scholar students with Django REST Framework, university-email-restricted auth, and Google Maps API for real-time GPS tracking and ride matching.
- Added safety features – SOS button, gender-specific filters, dual driver/rider mode — based on pain points surfaced through peer interviews.

NoteMate – AI Note-Taking Assistant

HEC GenAI Training Hackathon

- Built under hackathon conditions: students upload PDFs/notes, then query them using RAG, vector embeddings, and an LLM API, answers come from their own material, not the open web.
- Django REST backend handles upload, chunking, embedding generation, and retrieval end to end with no manual steps.

ANON – Decentralized Voting System

NUST Olympiad 2026 Hackathon

- P2P voting system using blind cryptographic signatures for anonymity and Byzantine Fault Tolerance so the system stays correct even with malicious nodes; DAG structure makes tampering detectable.

MediSearch – Symptom-Based Condition Search System

AI Course Project | FCCU

- Symptom checker that runs A*-style best-first search over a NetworkX knowledge graph edges weighted by symptom-condition co-occurrence frequency from 1,000 simulated student health profiles.
- Custom heuristic (edge weight x coverage ratio) prevents high-frequency conditions from drowning out specific matches; validated on overlapping profiles like fever+fatigue (flu vs. COVID) ranked correctly.
- Shipped as a Flask web app with autocomplete symptom input, top-5 ranked output, and a triage disclaimer.

PROGRAMMING SKILLS

Languages & Frameworks: Python, JavaScript, Django, Django REST Framework, Flask, ReactJS, NextJS, NodeJS, SQL

AI & Systems: RAG, LLM APIs, Vector Databases, NetworkX, Heuristic Search (A*/Best-First), Distributed Systems (P2P), Byzantine Fault Tolerance

Tools & Platforms: Git, GitHub, VS Code, PyCharm, Postman, Cursor, Google Maps API, SQLite, PostgreSQL

Languages: English (Fluent), Urdu (Native)